

James Ni

Email

jmni@berkeley.edu

GitHub

<https://github.com/jaynye>

ABOUT ME

I'm a postbaccalaureate researcher at Berkeley Artificial Intelligence Research (BAIR) since 2025, where I work closely with Professor Jitendra Malik, Professor Trevor Darrell, and Dr. Roei Herzig. I graduated with a B.E. in EECS with Highest Honors, and B.A. in Applied Mathematics, from the University of California, Berkeley. My research focuses on the long-term goal of building general-purpose humanoid robots. My current work specializes in learning humanoid locomotion, tracking and bimanual manipulation using deep reinforcement learning. I am eager to expand this work using task-agnostic representation learning, vision-based control, and test-time planning for long-horizon tasks. I believe that safe, practical generalization to novel contexts and environments is achieved through (i) visuomotor foundation models trained in (ii) simulation at scale, combined with (iii) vision-based planning using a learnt dynamics model of the world.

EDUCATION

2021-2025 | Bachelor of Engineering in Electrical Engineering and Computer Science (EECS), **Highest Honors**
Bachelor of Arts in Applied Mathematics, **Highest Distinction in General Scholarship**
University of California (UC), Berkeley

Selected Graduate Coursework: Computer Vision, Deep Reinforcement Learning (RL), Multi-agent Systems

Selected EECS Coursework: Machine Learning (A+), Robotic Manipulation (A+), Optimization Models, Computer Graphics, Databases, Network Architecture, Computer Security, Operating Systems, Algorithms, Digital Design and Integrated Circuits (ASIC)

Selected Mathematics Coursework: Complex Analysis, Abstract Algebra (A+), Abstract Linear Algebra (A+), Numerical Analysis, Stochastic Processes I-II (A+), Enumerative Combinatorics, Real Analysis

GPA: 4.0/4.0 overall

PUBLICATIONS

2025 | **J.M. Ni***, Z. Wang*, W. Lin*, A. Bar*, Y. Lecun, T. Darrell, J. Malik, R. Herzig.
From Generated Human Videos to Physically Plausible Robot Trajectories
In Review

2021 | **J.M. Ni**
Geometric Newton Inequalities
Proceedings of the American Mathematical Society

AWARDS & DISTINCTIONS

2025 | Arthur M. Hopkins Award, UC Berkeley

2021-2025 | Dean's List, UC Berkeley

PROFESSIONAL EXPERIENCE

2025-present | Postbaccalaureate Researcher
Berkeley Artificial Intelligence Research (BAIR), UC Berkeley
- Advised by Professor Jitendra Malik, Professor Trevor Darrell, and Dr. Roei Herzig
- Currently working on generalist humanoid control policies (confidential, in-progress)

2023-2025 | Undergraduate Researcher
BAIR, UC Berkeley
Advised by Professor Jitendra Malik, Professor Trevor Darrell, and Dr. Roei Herzig
Trained a 2D hand keypoint annotation model (ViT) for large-scale dataset generation using MMPose

CURRICULUM VITAE

Nov 15, 2025

	Trained humanoid locomotion, motion-capture based tracking, and bimanual pick and place policies using deep RL across 5 embodied modalities and 2 simulators, with sim-to-real transfer
2023-2025	Undergraduate Student Instructor UC Berkeley Undergraduate student instructor (uGSI) for CS 189, the flagship machine learning course, in Spring 2025 and Fall 2024, with duties including leading discussions, grading manager, hosting review sessions and office hours, exam and content creation Former uGSI for EECS 16B Lab in Spring 2024 Former lab assistant for EECS 16B Lab in Fall 2023 and Spring 2022

MAJOR PROJECTS

2025	Learning spades using multi-agent policy gradients - Implemented the high-dimensional, partially-observable, cooperative game of spades in a fully vectorized Gym env. - Use a System 1-System 2 hierarchy to decompose the bidding and playing phases of the game - Trained a playing module using multi-agent PPO which significantly outperforms random play
2025	Realistic lightning simulation - In collaboration with Jennifer Cao, Sonia Chacon, and Claire Ding - Implemented two models of realistic lightning simulation with shaders, real-time generation, visualization, a fully functioning GUI, and GPU acceleration - Created a data-efficient API for communicating between the application (C++) and the simulation backend (Python)
2024	Robust quadruped control, safety, and recovery using separating hyperplanes In collaboration with Kalie Ching, Harshika Jalan, and Akhil Vemuri Cast the simultaneous control, safety, and recovery of robotic systems as a hierarchical RL problem Extended original separating hyperplane work by Professor Jason Choi to a high-dimensional, under actuated system using RL at scale in IsaacGym, and showed successful recovery behavior in simulation
2023	Small-scale im2gps with masked autoencoders Created a challenging, small scale dataset for image geolocation based on an S2 cell partition of the USA Trained low-rank adaptation (LoRA) and decoder for predicting geolocation based on a pretrained masked autoencoder Achieved comparable results to historical state-of-the-art with very little compute (a single Google Colab instance)
2019-2021	Annotating RNA-seq single cell using NLP and GSEA - In collaboration with Alexander Wenzel and Jill Mesirov - Web-scraped gene descriptors from NCBI and used sentiment analysis with TF-IDF for feature extraction - Using gene set collections, aggregated significance and enrichment scores for keywords and clusters of cells using GESA - Allows researchers to automate the interpretation of single-cell data through the classification and description of cell clusters

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, Rust, Go, Javascript, HTML, Verilog
Technologies	CUDA, Docker, Linux, Slurm, Hammer VLSI, MongoDB, Git
Frameworks	PyTorch, IsaacGym, MuJoCo, JAX, Viser, OpenCV, ROS